

Audio lecture five. Early pregnancy loss, ectopic pregnancy, and recurrent pregnancy loss.

Companion reading handout for simultaneous listening.

This lecture is based on the American College of Obstetricians and Gynecologists, or A C O G, Practice Bulletin on Early Pregnancy Loss, the A C O G Practice Bulletin on Tubal Ectopic Pregnancy, Novak chapter 32 on early pregnancy loss and ectopic pregnancy, Novak chapter 33 on recurrent pregnancy loss, and Israeli Position Paper number 214 on recurrent pregnancy loss.

The clinical truth

Let us begin with a clinical truth.

Early pregnancy is biologically inefficient. Many conceptions do not become ongoing pregnancies. Some fail before the missed period. Some fail after implantation. Some implant outside the uterus. Some couples experience repeated losses. The clinician's job is to avoid two opposite errors: undertreating danger, especially ectopic pregnancy, and overtreating uncertainty, especially a potentially viable intrauterine pregnancy.

That balance shapes this entire lecture.

Early pregnancy loss

First, early pregnancy loss.

Early pregnancy loss means a nonviable intrauterine pregnancy in the first trimester, often defined as within **12 weeks and 6 days**. It may be an empty gestational sac or a gestational sac with an embryo or fetus without cardiac activity.

The most common cause is fetal chromosomal abnormality, especially aneuploidy. A C O G states that about **50 percent** of early pregnancy losses are due to fetal chromosomal abnormalities. Novak emphasizes that fetal aneuploidy is the most common cause of first trimester miscarriage, and recurrent pregnancy loss literature suggests at least **60 percent** of preclinical and early clinical losses are due to de novo fetal aneuploidy.

This is not just a statistic. It is a counseling principle. Most isolated early losses are not caused by something the patient did. They are usually the result of embryonic chromosomal error and biologic selection.

The risk rises with maternal age. Clinically recognized early pregnancy loss increases sharply after **35**, and by age **45** the risk is very high. This is because meiotic nondisjunction becomes more common with oocyte age.

Diagnosis

Now diagnosis.

Symptoms such as vaginal bleeding and cramping are common in normal pregnancy, early loss, ectopic pregnancy, and molar pregnancy. Therefore, treatment should not begin until diagnosis is secure. A C O G emphasizes that treating early pregnancy loss before confirmation can interrupt a normal pregnancy, create complications, or expose an ongoing pregnancy to harmful agents.

Ultrasound and serial beta human chorionic gonadotropin, spoken beta H C G, are the core tools.

The diagnostic standard has become deliberately conservative. For example, thresholds for crown rump length without cardiac activity and mean sac diameter without embryo are designed to avoid false diagnosis of nonviability. The philosophy is more important than the exact numbers for this lecture: **when uncertain, repeat imaging rather than intervene too soon, if the patient is stable.**

Management of confirmed early pregnancy loss

Management of confirmed early pregnancy loss has three main pathways: expectant, medical, and surgical.

Expectant management

Expectant management can be effective, especially in first trimester incomplete loss. A C O G reports success in about **80 percent** with adequate time, up to **8 weeks**. It is less suitable later in pregnancy because hemorrhage risk rises.

Medical management

Medical management usually uses misoprostol, often with mifepristone in settings where available and indicated. It shortens time to expulsion and reduces need for surgical evacuation.

Surgical management

Surgical management, usually suction curettage or vacuum aspiration, is appropriate for heavy bleeding, infection, hemodynamic instability, patient preference, or when rapid completion is needed. Suction is preferred over sharp curettage alone.

Rh D immune globulin should be considered for Rh negative patients, especially later in the first trimester and particularly with surgical management.

Early pregnancy loss summary

Pause and summarize early pregnancy loss.

The first priority is diagnostic certainty. Do not treat a desired pregnancy as a loss until the ultrasound and beta H C G picture is secure. Once confirmed, expectant, medical, and surgical management are all valid, and the choice depends on stability, infection, bleeding, timing, and patient preference.

Ectopic pregnancy

Now let us transition to ectopic pregnancy.

Ectopic pregnancy is implantation outside the uterine cavity. More than **90 percent** are tubal. It remains dangerous because rupture can cause hemorrhage and maternal death. A C O G notes that ectopic pregnancy accounts for about **2 percent** of reported pregnancies, and ruptured ectopic pregnancy remains a major cause of pregnancy-related hemorrhagic mortality.

The classic symptom triad, from Novak, is **pain, amenorrhea, and vaginal bleeding**. This triad is present in about half of patients and is more typical of rupture. Do not wait for the full triad. Every reproductive-aged patient with abdominal pain or vaginal bleeding should have pregnancy status assessed.

Risk factors include prior ectopic pregnancy, tubal damage, pelvic infection, prior tubal surgery, infertility, assisted reproduction, smoking, and age over **35**. But one half of patients have no known risk factor. That is the humbling part. Absence of risk factors does not exclude ectopic pregnancy.

Ectopic pregnancy diagnosis

Diagnosis begins with transvaginal ultrasound and confirmation of pregnancy.

A definitive intrauterine pregnancy, meaning a gestational sac with yolk sac or embryo in the uterus, usually excludes ectopic pregnancy except in rare heterotopic pregnancy. A definitive ectopic pregnancy is an extrauterine gestational sac with yolk sac or embryo. But most cases are not that clean initially.

Pregnancy of unknown location

This brings us to pregnancy of unknown location.

Pregnancy of unknown location is not a diagnosis. It is a temporary state. The pregnancy test is positive, but ultrasound does not yet show a definite intrauterine or ectopic pregnancy. Stable patients who desire pregnancy continuation should be followed with repeat ultrasound and serial beta H C G until location and viability are clear.

The discriminatory zone

Now, the discriminatory zone.

The discriminatory level is the beta H C G value above which an intrauterine pregnancy should usually be visible by transvaginal ultrasound. Historically, lower values were used. But A C O G warns that the concept can be dangerous if used too aggressively. Some viable intrauterine pregnancies are not seen even when beta H C G is above traditional thresholds. Therefore, if the discriminatory level is used, it should be conservatively high, as high as **3500 milli international units per milliliter**, to avoid interrupting a desired intrauterine pregnancy.

That is a high-yield exam point, but it is also a patient safety point. A single beta H C G should not diagnose ectopic pregnancy by itself. It must be interpreted with symptoms, ultrasound, gestational age, and serial change.

Serial beta H C G patterns help. In a viable intrauterine pregnancy, beta H C G should rise over 48 hours, but the minimal expected rise depends on the starting value. Lower starting values rise faster; higher starting values rise more slowly. Falling values suggest a failing pregnancy, but ectopic rupture can still occur at low or falling levels. So follow-up continues until resolution or diagnosis.

Management of ectopic pregnancy

Now management of ectopic pregnancy.

If the patient is unstable, has signs of rupture, significant intraperitoneal bleeding, or an acute abdomen, this is surgical. Do not delay for perfect lab patterns.

If the patient is stable and the ectopic is unruptured, treatment can be medical or surgical depending on findings, patient preference, follow-up reliability, beta H C G level, mass size, cardiac activity, and contraindications.

Methotrexate

Methotrexate is the classic medical treatment. It is a folate antagonist that inhibits rapidly dividing trophoblastic tissue. Candidates must be hemodynamically stable,

with confirmed or high suspicion ectopic pregnancy, no rupture, ability to follow up, and no absolute contraindication.

Absolute contraindications include intrauterine pregnancy, hemodynamic instability, ruptured ectopic pregnancy, breastfeeding, immunodeficiency, significant anemia or leukopenia or thrombocytopenia, active pulmonary disease, active peptic ulcer disease, clinically important hepatic disease, and clinically important renal dysfunction. The test answer in the guide highlights renal insufficiency.

Relative contraindications include high initial beta H C G, commonly over **5000**, ectopic mass greater than **4 centimeters**, and embryonic cardiac activity. These predict lower success, not absolute impossibility.

After methotrexate, beta H C G is followed. In the single-dose regimen, failure to fall by at least **15 percent from day 4 to day 7** suggests treatment failure and need for additional methotrexate or surgery.

Surgery

Surgery is usually laparoscopy when feasible. Salpingectomy removes the affected tube. Salpingostomy removes the ectopic pregnancy and leaves the tube. The choice depends on fertility goals, tube damage, bleeding, contralateral tube status, and recurrence risk. If salpingostomy is performed, beta H C G must be followed because persistent trophoblast can remain.

Ectopic pregnancy summary

Pause and summarize ectopic pregnancy.

Think ectopic in any reproductive-aged patient with pain or bleeding. The classic triad is pain, amenorrhea, and vaginal bleeding, but it is absent in many. Do not diagnose by one beta H C G. Use ultrasound, serial beta H C G, symptoms, and clinical risk. Use a conservative discriminatory level, up to **3500**, if using it at all. Methotrexate is for stable unruptured cases without contraindications, and renal dysfunction is a contraindication.

Recurrent pregnancy loss

Now we transition to recurrent pregnancy loss, or R P L.

Recurrent pregnancy loss is emotionally devastating and biologically complex. Traditional definitions use three or more clinical losses, but evaluation may begin after two consecutive losses because the prevalence of abnormal findings is similar after two versus three losses.

Novak gives the first counseling anchor: even after recurrent losses, most patients are more likely to have a live birth in the next pregnancy than another miscarriage.

That does not minimize the loss. It frames prognosis honestly.

Etiology

Now etiology.

The most common cause of first trimester miscarriage remains fetal aneuploidy. In recurrent loss, about **50 percent** of couples may have no specific abnormality after evaluation. Identified categories include genetic, anatomic, endocrine, antiphospholipid syndrome, and other factors.

Genetic evaluation

Genetic evaluation has two sides.

The pregnancy tissue can be tested, preferably with chromosomal microarray when available, because it does not require culture and can identify many numerical and structural abnormalities. Israeli Position Paper number 214 notes that chromosomal abnormalities are expected in **50 to 60 percent** of miscarriages, mostly sporadic.

The parents can be tested with karyotypes to detect balanced rearrangements, such as reciprocal or Robertsonian translocations. Novak estimates parental balanced structural rearrangements in about **3 to 5 percent** of recurrent pregnancy loss. The key concept is balance. A parent may be healthy because no genetic material is gained or lost, but gametes can be unbalanced, leading to miscarriage or affected offspring. Microarray cannot detect a balanced rearrangement in the parents, because the total copy number is normal. Standard karyotype is needed.

Anatomic evaluation

Anatomic evaluation matters because implantation requires a receptive, vascularized uterine cavity. Septum, adhesions, submucosal fibroids, polyps, and congenital anomalies can impair implantation or placentation. Novak notes that anatomic issues account for roughly **10 to 12 percent** in many studies, and Israeli Position Paper number 214 states that acquired and congenital anatomic defects are found in about **15 percent**.

Evaluation can include hysterosalpingography, saline sonography, three dimensional ultrasound, or office hysteroscopy. The answer key flagged hysteroscopy or uterine cavity evaluation as likely highest yield if the question asks which test has high practical yield.

Endocrine evaluation

Endocrine evaluation focuses on diabetes, thyroid disease, and prolactin. Poorly controlled diabetes increases miscarriage risk and congenital anomalies; well-controlled diabetes does not appear to increase recurrent loss risk. Israeli Position Paper number 214 specifically notes the importance of hemoglobin A 1 c below **6 percent** in diabetic patients. Thyroid dysfunction and hyperprolactinemia are treatable and should be sought.

Antiphospholipid syndrome

Now antiphospholipid syndrome, or A P S.

A P S is the classic treatable immune-thrombotic cause of recurrent pregnancy loss. Testing includes lupus anticoagulant, anticardiolipin IgG and IgM, and anti beta 2 glycoprotein 1 IgG and IgM. Diagnosis requires clinical criteria and persistent laboratory positivity. Mechanistically, antiphospholipid antibodies may interfere with trophoblast function, placentation, coagulation balance, and placental vascular integrity.

Treatment usually involves low-dose aspirin and heparin in patients who meet criteria, and live birth rates improve substantially compared with untreated historical outcomes.

Inherited thrombophilias

Now, inherited thrombophilias.

This is where older practice and current evidence diverge. It is biologically tempting to test factor V Leiden, prothrombin mutation, protein C, protein S, antithrombin, homocysteine, and methylene tetrahydrofolate reductase, or M T H F R. But Novak states that A C O G and the American Society for Reproductive Medicine, or A S R M, do not recommend routine inherited thrombophilia or M T H F R screening for recurrent pregnancy loss unless there is personal or strong family history of thrombosis. The reason is that evidence does not show consistent benefit from routine screening or anticoagulation in unexplained recurrent loss.

So the memorable contrast is: **test for A P S; do not reflexively test inherited thrombophilias without thrombosis context.**

Treatment and follow-up

Now treatment and follow-up.

Treatment is cause-specific when a cause is found: genetic counseling for parental rearrangements, correction of significant uterine cavity abnormalities, control of thyroid disease, prolactin, and diabetes, aspirin and heparin for A P S. Progesterone

may benefit selected patients with recurrent losses, but broad use remains nuanced. Immunotherapies such as paternal leukocyte immunization, broad immunosuppression, and intravenous immunoglobulin remain controversial and are not routine.

Equally important is supportive care. Novak emphasizes close monitoring with beta H C G levels and ultrasound, psychological support, and karyotypic analysis of tissue from subsequent miscarriage. This matters because uncertainty itself causes harm. Early ultrasound confirmation, clear follow-up plans, and compassionate counseling are part of therapy.

Recurrent pregnancy loss summary

Pause and summarize recurrent pregnancy loss.

The practical workup is focused, not endless. Look for uterine cavity problems, parental chromosomal rearrangements, chromosomal abnormalities in pregnancy tissue, thyroid disease, diabetes, prolactin problems, and A P S. Do not reflexively order inherited thrombophilia panels or M T H F R testing unless the thrombosis history justifies it.

Final synthesis

Now final synthesis.

1. **First, early pregnancy loss is common, and the most common cause is fetal chromosomal abnormality.** Counsel with biology, not blame.
2. **Second, ectopic pregnancy is the dangerous mimic.** The classic triad is pain, amenorrhea, and bleeding, but many patients do not present classically. Use ultrasound and serial beta H C G, and be conservative with the discriminatory level.
3. **Third, methotrexate is for stable, unruptured ectopic pregnancy when follow-up is reliable and contraindications are absent.** Clinically important renal dysfunction is a contraindication.
4. **Fourth, recurrent pregnancy loss evaluation should be focused: uterine anatomy, parental and pregnancy genetics, endocrine disease, and A P S.** Routine inherited thrombophilia testing is usually not supported without thrombosis history.
5. **Finally, remember the emotional frame.** In early pregnancy, your precision protects desired pregnancies from premature intervention, and your vigilance protects patients from ectopic rupture.

End of lecture five.